The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery

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One of the grand challenges of artificial general intelligence is developing agents capable of conducting scientific research and discovering new knowledge. While frontier models have already been used as aides to human scientists, e.g. for brainstorming ideas, writing code, or prediction tasks, they still conduct only a small part of the scientific process. This paper presents the first comprehensive framework for fully *automatic scientific discovery*, enabling frontier large language models (LLMs) to perform research independently and communicate their findings. We introduce The AI Scientist, which generates novel research ideas, writes code, executes experiments, visualizes results, describes its findings by writing a full scientific paper, and then runs a simulated review process for evaluation. In principle, this process can be repeated to iteratively develop ideas in an open-ended fashion and add them to a growing archive of knowledge, acting like the human scientific community. We demonstrate the versatility of this approach by applying it to three distinct subfields of machine learning: diffusion modeling, transformer-based language modeling, and learning dynamics. Each idea is implemented and developed into a full paper at a meager cost of less than $15 per paper, illustrating the potential for our framework to democratize research and significantly accelerate scientific progress. To evaluate the generated papers, we design and validate an automated reviewer, which we show achieves near-human performance in evaluating paper scores. The AI Scientist can produce papers that exceed the acceptance threshold at a top machine learning conference as judged by our automated reviewer. This approach signifies the beginning of a new era in scientific discovery in machine learning: bringing the transformative benefits of AI agents to the *entire* research process of AI itself, and taking us closer to a world where *endless affordable creativity and innovation* can be unleashed on the world's most challenging problems. Our code is open-sourced at https://github.com/SakanaAI/AI-Scientist.

1. Introduction

The modern scientific method (Chalmers, 2013; Dewey, 1910; Jevons, 1877) is arguably one of the greatest achievements of the Enlightenment. Traditionally, a human researcher collects background knowledge, drafts a set of plausible hypotheses to test, constructs an evaluation procedure, collects evidence for the different hypotheses, and finally assesses and communicates their findings. Afterward, the resulting manuscript undergoes peer review and subsequent iterations of refinement. This procedure has led to countless breakthroughs in science and technology, improving human quality of life. However, this iterative process is inherently limited by human researchers' ingenuity, background knowledge, and finite time. Attempting to automate general scientific discovery (Langley, 1987, 2024; Waltz and Buchanan, 2009) has been a long ambition of the community since at least the early 70s, with computer-assisted works like the Automated Mathematician (Lenat, 1977; Lenat and Brown, 1984) and DENDRAL (Buchanan and Feigenbaum, 1981). In the field of AI, researchers have envisioned the possibility of automating AI research using AI itself (Ghahramani, 2015; Schmidhuber, 1991, 2010a,b, 2012), leading to "AI-generating algorithms" (Clune, 2019). More recently, foundation models have seen tremendous advances in their general capabilities (Anthropic, 2024; Google DeepMind Gemini Team, 2023; Llama Team, 2024; OpenAI, 2023), but they have only been shown to accelerate individual parts of the research pipeline, e.g. the writing of scientific manuscripts (Altmäe et al., 2023;

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Dinu et al., 2024; Ifargan et al., 2024; Majumder et al., 2024), as a muse to brainstorm ideas (Baek et al., 2024; Girotra et al., 2023; Wang et al., 2024b), or aides to coding (Gauthier, 2024). To date, the community has yet to show the possibility of executing entire research endeavors without human involvement.

Traditional approaches to automating research projects have so far relied on carefully constraining the search space of potential discoveries, which severely limits the scope of exploration and requires substantial human expertise and design. For example, significant advancements in materials discovery (Merchant et al., 2023; Pyzer-Knapp et al., 2022; Szymanski et al., 2023) and synthetic biology (Hayes et al., 2024; Jumper et al., 2021) have been achieved by restricting exploration to well-characterized domains with predefined parameters, which allows for targeted progress but limits broader, open-ended discovery and addressing only a subset of the scientific process, without encompassing tasks such as manuscript preparation. Within the field of machine learning itself, research automation has largely been restricted to hyperparameter and architecture search (He et al., 2021; Hutter et al., 2019; Lu et al., 2022b; Wan et al., 2021, 2022) or algorithm discovery (Alet et al., 2020; Chen et al., 2024b; Kirsch et al., 2019; Lange et al., 2023a,b; Lu et al., 2022a; Metz et al., 2022) within a hand-crafted search space. Recent advances in LLMs have shown the potential to extend the search space to more generalized, code-level solutions (Faldor et al., 2024; Lehman et al., 2022; Lu et al., 2024a; Ma et al., 2023). However, these approaches remain constrained by rigorously-defined search spaces and objectives, which limit the breadth and depth of possible discoveries.

In this paper, we introduce The AI Scientist, the first fully automated and scalable pipeline for end-to-end paper generation, enabled by recent advances in foundation models. Given a broad research direction and a simple initial codebase, The AI Scientist seamlessly performs ideation, a literature search, experiment planning, experiment iterations, manuscript writing, and peer reviewing to produce insightful papers. Furthermore, in principle The AI Scientist can run in an open-ended loop, building on its previous scientific discoveries to improve the next generation of ideas. This allows us to speed up the slow nature of scientific iteration at a surprisingly low financial cost (~$15/paper) and represents a step towards turning the world's ever-increasing computing resources into the scientific breakthroughs needed to tackle the core challenges of the 21st century. Here, we focus on Machine Learning (ML) applications, but this approach can more generally be applied to almost any other discipline, e.g. biology or physics, given an adequate way of automatically executing experiments (Arnold, 2022; Kehoe et al., 2015; Zucchelli et al., 2021).

By leveraging modern LLM frameworks like chain-of-thought (Wei et al., 2022) and self-reflection (Shinn et al., 2024) to improve decision-making, The AI Scientist is able to generate its own scientific ideas and hypotheses, as well as a plan for testing them with experiments. Next, The AI Scientist implements plan-directed code-level changes to the experiment "template" using the state-of-the-art coding assistant Aider (Gauthier, 2024), and executes experiments to collect a set of computational results, which are in turn used to draft a scientific paper. The AI Scientist then performs an automated paper-reviewing process using guidelines from a standard machine learning conference. Finally, The AI Scientist adds the completed ideas and reviewer feedback to its archive of scientific findings, and the process repeats. Crucially, the generated paper and experimental artifacts The AI Scientist produces allow us to easily interpret and judge its findings post-hoc, allowing human scientists to also benefit from what is learned.

Our contributions are summarized as follows:

1. We introduce the first end-to-end framework for fully automated scientific discovery in Machine Learning research, enabled by frontier LLMs (Section 3). This fully automated process includes idea generation, experiment design, execution, and visualizing and writing up the results into a full manuscript.

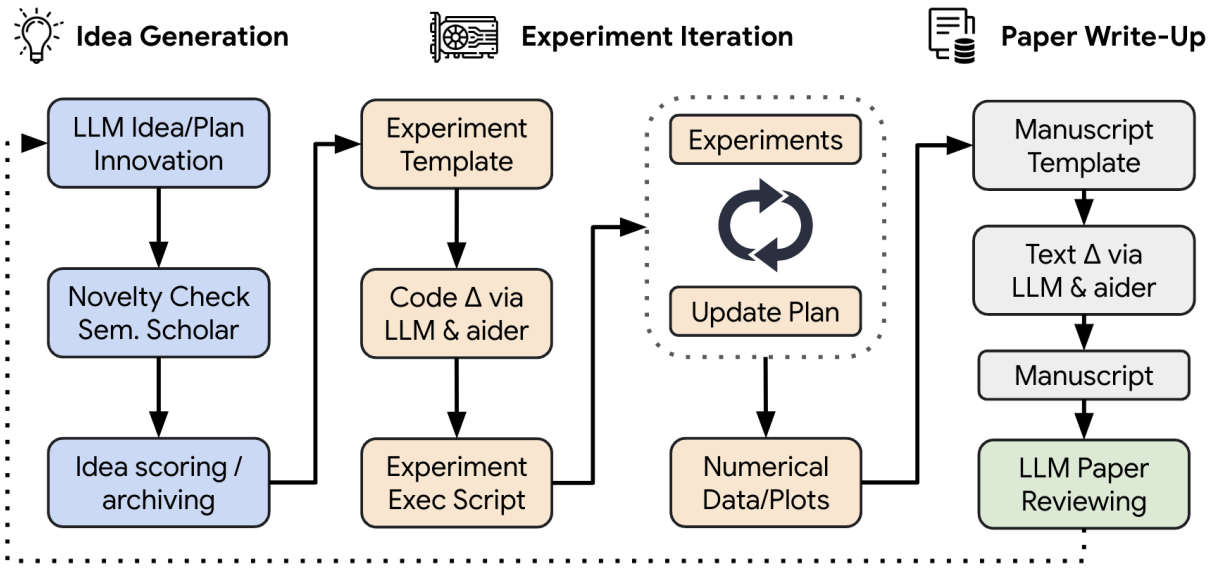


Figure 1 | Conceptual illustration of THE AI SCIENTIST, an end-to-end LLM-driven scientific discovery process. THE AI SCIENTIST first invents and assesses the novelty of a set of ideas. It then determines how to test the hypotheses, including writing the necessary code by editing a codebase powered by recent advances in automated code generation. Afterward, the experiments are automatically executed to collect a set of results consisting of both numerical scores and visual summaries (e.g. plots or tables). The results are motivated, explained, and summarized in a LaTeX report. Finally, THE AI SCIENTIST generates an automated review, according to current practice at standard machine learning conferences. The review can be used to either improve the project or as feedback to future generations for open-ended scientific discovery.

2. To assess the quality of the generated papers, we introduce a foundation model-based reviewing process in Section 4. This process achieves near-human-level performance across multiple evaluation metrics (e.g. 65% vs. 66% balanced accuracy) when evaluated on ICLR 2022 OpenReview data. The reviews further enable THE AI SCIENTIST to select the best ideas for "publication" to an ever-growing archive of scientific discoveries, and the process can be repeated to build on these discoveries, just as in the human scientific community.
3. THE AI SCIENTIST can generate hundreds of interesting, medium-quality papers over the course of a week. In this report, we focus on a subset of these papers, highlighting novel insights in diffusion modeling, language modeling, and grokking. We perform an in-depth case study into one selected paper in Section 5, and present aggregate results in Section 6.
4. We conclude the paper with an extensive discussion on the limitations, ethical considerations, and future outlook of our approach in Sections 8 and 9.

2. Background

Large Language Models. In this paper, we build our automated scientist from autoregressive large language models (LLMs, Anthropic (2023); Google DeepMind Gemini Team (2023); Llama Team (2024); OpenAI (2023); Zhu et al. (2024)) which learn to generate text completions by modeling the conditional probability of a new token (similar to a word) given the preceding tokens, $p(x_t|x_{<t};\theta)$, and sampling at test-time. Together with vast data and model scaling, this enables LLMs to not only generate coherent text, but crucially also exhibit human-like abilities, including commonsense knowledge (Talmor et al., 2019), reasoning (Wei et al., 2022), and the ability to write code (Chen et al., 2021; Xu et al., 2022).

LLM Agent Frameworks. Typical applications of LLMs often involve embedding the model into an "agent" (Wang et al., 2024a) framework, including the following possibilities: the structuring of